Quantum error mitigation by layerwise Richardson extrapolation

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Introduction

Problem: Quantum computers are susceptible to environmental noise.

QEM: Quantum error mitigation (QEM) is accessible today to deal with noise.

In this work, we:

- Run experiments on hardware and apply QEM.
- Analyze how these techniques perform on different devices.

Layerwise folding

Unmitigated: Evaluation of an unmitigated expectation value A' is a *trial*. Perform t trials $A'_{[1]}, ..., A'_{[t]}$. Quantify the estimation error via the root-mean-square error (RMSE)

$$\sqrt{\frac{1}{t}\sum_{i=1}^{t}\left(A'_{[i]} - A\right)^2}.$$

Mitigated: Evaluation of A_{QEM} as a *QEM trial*. Perform t QEM trials $A^{[1]}_{\mathrm{QEM}}, ..., A^{[t]}_{\mathrm{QEM}}$ and evaluate RMSE

$$\sqrt{\frac{1}{t}\sum_{i=1}^{t}\left(A^{[i]}_{\mathrm{QEM}} - A\right)^2}.$$

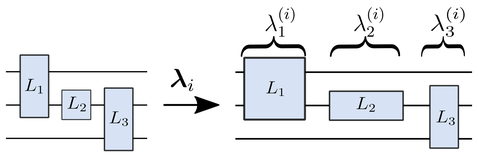

Figure 1: Improvement factor results for $n = 3$ qubit experiments (d is Clifford depth).

Quantum error mitigation

ZNE: We evaluate $k_{\mathrm{ZNE}} = 3$ noisy expectation values $A'(\lambda_i)$ at different noise scale factors $\lambda_i \in \{1, 2, 3\}$. Zero noise limit is obtained as:

$$A_{\mathrm{ZNE}} = \sum_{i=1}^{k_{\mathrm{ZNE}}} \eta_i A'(\lambda_i)$$

We consider both *linear* and *Richardson* extrapolation. The best fit coefficients η_i are

Methodology Overview

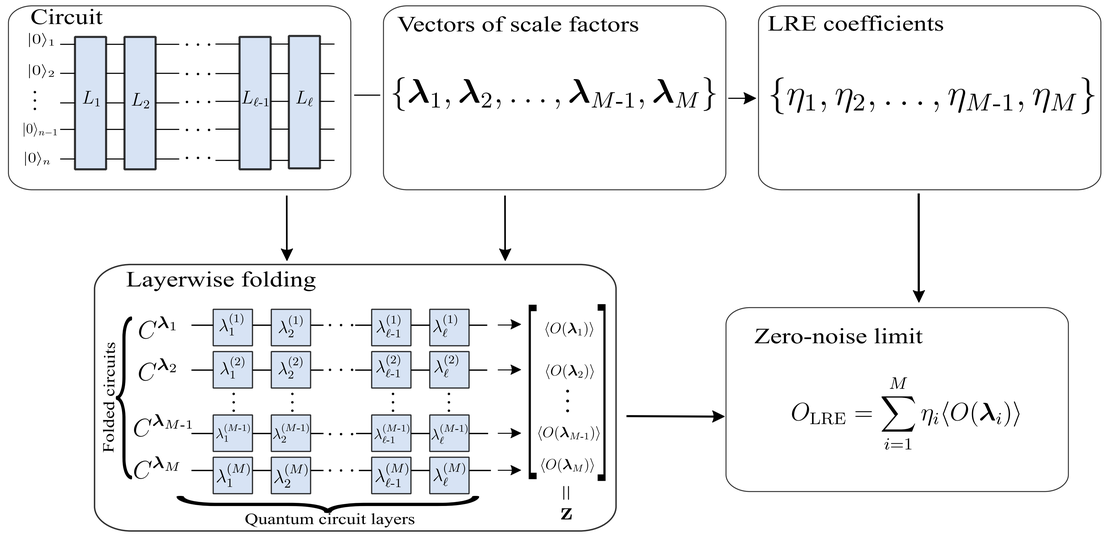

Figure 2: LRE experimental workflow. Input: n-qubit quantum circuit with ℓ layers. Given the parameter ℓ and the extrapolation order d, we generate $M = \binom{d+\ell}{d}$ linearly-independent vectors of scale factors. We perform layerwise folding on the input circuit generating M different circuits, one for each vector of scale factors. For each resulting circuit, we estimate the respective expectation value. The linear combination coefficients $\{\eta_j\}$ can be computed straightforwardly from the multivariate Lagrange interpolation formula and only depend on the scale factor vectors. By taking a linear combination of the noise-scaled expectation values, we obtain the error-mitigated result.

Improvement factor

Improvement factor: Quantify the performance of quantum error mitigation:

$$\mu_{\mathrm{QEM}} := \frac{\sqrt{N \sum_{C\in\mathcal{C},\hat{A}\in\hat{\mathcal{A}}} \sum_{i=1}^{t} (A'_{[i]} - A)^2}}{\sqrt{N_{\mathrm{QEM}} \sum_{C\in\mathcal{C},\hat{A}\in\hat{\mathcal{A}}} \sum_{i=1}^{t} (A^{[i]}_{\mathrm{QEM}} - A)^2}}$$

the circuit C is implicit in the expectation values A, $A'_{[i]}$, and $A^{[i]}_{\mathrm{QEM}}$, e.g. $A = \mathrm{tr}[C|0\rangle\langle 0|C^\dagger \hat{A}]$.

Results: Applied ZNE(L), ZNE(R), and PEC to RB circuit and mirror circuit benchmarks on IBM, IonQ, and Rigetti quantum computers. For each of these experiments, we compute the improvement factor.

Results

Takeaway: "Out-of-the-box" QEM with Mitiq improves (on average) measurement results, (even when normalizing by additional resources).

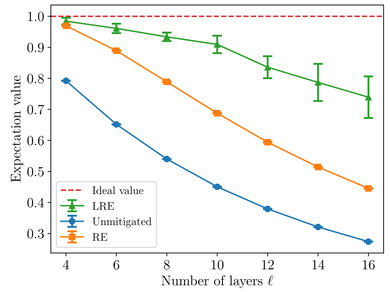

Figure 3: (Bottom) Unmitigated expectation values (orange) and corresponding mitigated expectation values using ZNE(R) (green) for $n = 3$ qubit RB circuits executed on IBMQ Lima. (Top) Improvement factor at each depth.

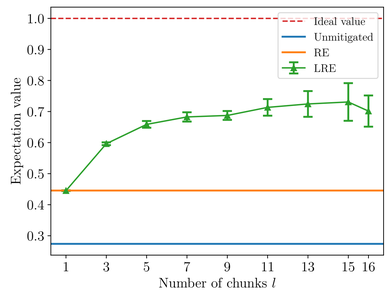

Figure 4: Improvement factor results for n = 12 qubit experiments on IBMQ Kolkata device (d is Clifford depth, corresponding to 100 two-qubit and 600 single-qubit gates).

Software

We implemented our experiments in Python 3.11 using Mitiq v.0.31.0. Supporting code is available at [https://github.com/unitaryfund/research/].

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